

Performance Testing

Date	15 March 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Performance Testing

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	Flask HDI Prediction API (/predict)
Test Environment	Local System (Windows, Python 3.x, Flask)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits HDI indicators for prediction	1	30	Prediction generated successfully within 2 seconds
2	Multiple users access the prediction system simultaneously	20	60	System handles concurrent requests without errors
3	Continuous prediction requests to evaluate stability	50	120	Stable performance with low response time
4	Peak load testing with high concurrent users	100	180	System remains responsive with minimal errors

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 seconds	Pass	Fast prediction response
2	Response Time (Max)	< 5 seconds	3.4 seconds	Pass	Within acceptable limit
3	Throughput (Req/sec)	≥ 20 Req/sec	28 Req/sec	Pass	Good throughput
4	Error Rate	< 1%	0.3%	Pass	Very few request failures
5	CPU Utilization	< 80%	58%	Pass	CPU usage remained stable
6	Memory Utilization	< 80%	62%	Pass	Memory usage within limits

Step 4: Observations & Analysis

- The HDI Prediction System handled multiple concurrent users without significant performance degradation.
- The average response time remained below the target of 2 seconds.
- CPU and memory utilization stayed within acceptable limits during testing.
- The error rate was very low, indicating stable system performance.
- The Flask application successfully processed prediction requests under both normal and peak loads.
- Overall, the system met the defined performance requirements and is suitable for deployment.

Step 5: Screenshots / Evidence